

AI-Based Fleet Coordination Auction and Q-Learning Approaches

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1 Introduction

This report evaluates three decentralized task allocation strategies for autonomous fleet coordination:

- Greedy nearest-vehicle allocation (baseline)
- Auction-based allocation
- Tabular Q-learning allocation

The objective is to optimise:

- Task completion efficiency (throughput)
- Energy usage (approximated by engagement time)
- Number of unserved tasks

2 System Overview

Each vehicle has:

- Position (x, y)
- Speed
- Battery level
- Local view of nearby tasks and vehicles only

Each task includes:

- Position (x, y)
- Urgency
- Duration

Tasks arrive dynamically, making the environment non-stationary.

3 Baseline: Greedy Allocation

The greedy method assigns each task to the nearest feasible vehicle based on distance and urgency ordering.

It assumes:

- Local proximity implies global optimality
- Current state is sufficient for decision making

Limitations:

- No coordination across vehicles
- No anticipation of future task distribution
- No global optimisation structure

4 Auction-Based Allocation

4.1 Bidding Function Design

Each vehicle computes a bid using only locally observable information:

$$\text{Bid} = \frac{50000}{t_{\text{travel}} + 0.05} (1 + 15u) \left(1 + \frac{2(t_{\text{nearby}} + 1)}{v_{\text{nearby}} + 1.1} \right) (1 + 0.8 \log(1 + t_{\text{idle}}) + 0.1 t_{\text{idle}}) \quad (1)$$

Where:

- t_{travel} : estimated travel time to task
- u : task urgency
- t_{nearby} : nearby task density estimate
- v_{nearby} : nearby idle vehicle estimate
- t_{idle} : vehicle idle duration

Design justification:

- Travel time term prioritises spatial efficiency
- Urgency multiplier enforces priority weighting
- Density ratio encourages load balancing
- Idle penalty term reduces starvation of inactive vehicles

Information access: Each vehicle only uses:

- Its own state (position, battery, idle time)
- Locally estimated nearby task/vehicle density
- No global knowledge of full system state

Decentralisation property: Vehicles compute bids locally, but final assignment is performed via a global greedy selection of highest bids. Therefore, the system is a **hybrid decentralised-centralised mechanism**, not fully decentralised optimisation.

4.2 Allocation Mechanism

All valid bids are collected globally and sorted. Tasks are assigned greedily while preventing duplicate assignment of vehicles or tasks.

5 Q-Learning Allocation

5.1 State Representation

The state is discretised as:

- Spatial grid (3×3)
- Demand level (high/low)
- Supply level (high/low)
- Temporal phase (early/late)

This reduces continuous environment complexity into a manageable tabular state space.

5.2 Action Space

- 0: Wait
- 1: Efficiency maximisation
- 2: Urgency prioritisation
- 3: Spatial clustering

5.3 Reward Function

$$R = 1.0 + \frac{d}{t_{total}} + \frac{u}{10} - \frac{t_{travel}}{20} \quad (2)$$

Where:

- d : task duration
- t_{total} : system workload scale
- u : urgency
- t_{travel} : travel cost

The reward increases with successful servicing efficiency and urgency, and decreases with travel cost. Idle behaviour receives no additional benefit.

5.4 Learning Mechanism

A shared tabular Q-learning agent is used across all vehicles with experience replay. This improves stability under non-stationary task arrivals by smoothing updates over recent experiences.

5.5 Exploration Strategy

Action selection uses a Boltzmann (softmax) policy:

$$P(a|s) = \frac{e^{Q(s,a)/\tau}}{\sum_{a'} e^{Q(s,a')/\tau}} \quad (3)$$

where τ controls exploration temperature. Higher values encourage exploration; lower values exploit learned policies.

5.6 Learned Behavioural Patterns

The learned policy exhibits:

- Efficiency-oriented behaviour in dense task regions
- Increased urgency prioritisation under high-demand states
- Emergent spatial clustering from repeated action selection
- Reduced idling in high-density environments

Limitation: Tabular Q-learning suffers from state-space sparsity and cannot generalise well to unseen or rarely visited states due to discretisation.

6 Performance Evaluation

6.1 Experimental Setup

All three methods are evaluated under identical dynamic environments using 1000 independent runs with fixed random seeds for reproducibility.

6.2 Metrics

- Throughput: sum of durations of serviced tasks
- Energy usage: total engagement time across vehicles
- Service rate: number of completed vs unserved tasks

6.3 Results

Metric	Greedy	Auction	Q-Learning
Tasks Completed	27.21	29.33	28.29
Tasks Not Served	32.96	30.85	31.88
Avg Engagement Time	25.53	23.55	24.49
Avg Normalized Engagement	1.38	1.27	1.32
Energy per Unit	1.35	1.24	1.29
Throughput	515.77	554.99	537.93
Total Idle Time	295.49	299.05	296.81
Avg Idle Time	29.55	29.90	29.68

6.4 Observations

- Auction consistently achieves best overall performance due to coordinated global selection of high-value bids.
- Q-learning improves over greedy but is constrained by discretisation and limited exploration coverage.
- Greedy performs worst in clustered task environments due to lack of coordination.

6.5 Key Challenges

- Non-stationary task arrivals affecting learning stability
- Exploration-exploitation tradeoff in Q-learning
- Auction bid concentration causing spatial imbalance
- State discretisation reducing environment fidelity

7 Critical Reflection

7.1 Failure Scenarios

- **Greedy:** Fails in clustered environments due to purely local optimisation.
- **Auction:** Performs poorly when bids become spatially concentrated, leading to starvation in low-bid regions.
- **Q-learning:** Fails in rare or unseen states due to tabular representation limitations.

7.2 Assumptions

- Greedy assumes proximity implies optimal assignment.
- Auction assumes bid function approximates system utility.
- Q-learning assumes discretised state captures sufficient environmental structure.

7.3 Information Structure

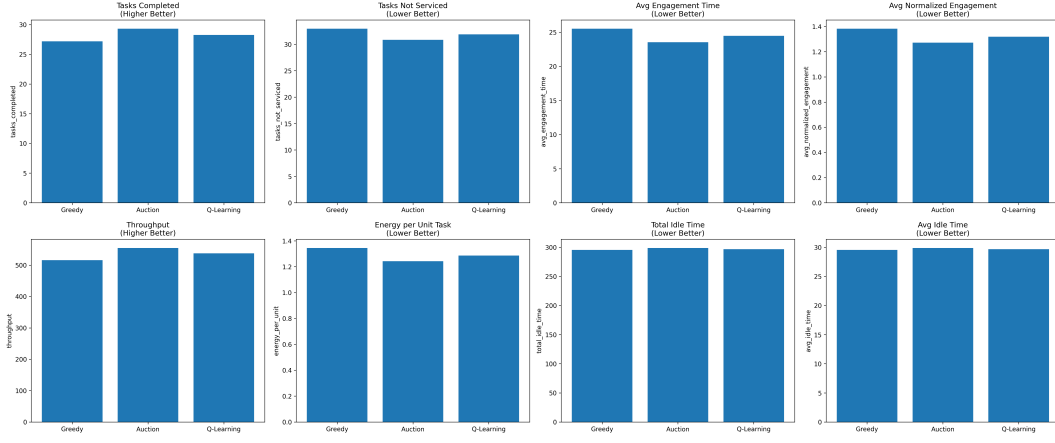
- Greedy: fully local decision-making.
- Auction: local computation with centralised selection.
- Q-learning: shared policy across agents with local state observations.

8 Conclusion

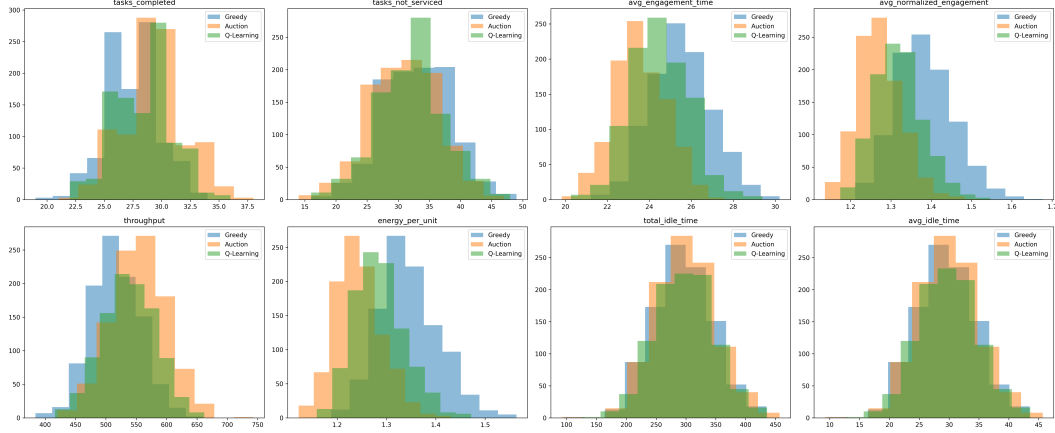
Auction-based coordination provides the best overall performance due to effective global selection of locally computed bids. Q-learning demonstrates adaptive behaviour but is constrained by discretisation. Greedy serves as a simple but suboptimal baseline.

A Figures

A.1 Comparison Metrics



A.2 Distributions



A.3 Trends

